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Proposal for:

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Principal
BR+A
105 Madison Ave, 10th floor
New York, NY
10016

Proposal Date:

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Proposal Number:

QU-0075

Reference

US NY BR+A Student Life Center

Understanding of Project

The project is a confidential site in north-western Manhattan (Morningside Heights neighborhood). The building will be a 200,000 ft² student life building for a confidential university. The total plant for the building is anticipated to be approximately 500 tons cooling and 2.5 million Btu/hr heating. BR+A has requested design assistance for the conceptual design and stamped construction documents for the ground heat exchanger (GHX) up to and including the manifold.

Proposed Design Methodology

Phase 1: Feasibility of GSHP System

The size, cost, and performance of a GHX is sensitive to the peak heat transfer to and from the ground. It is also sensitive to the total annual heat rejection to the ground when the building is being cooled compared to the amount of heat extracted from the ground to heat the building. This is referred to as the annual energy balance.

Accurate hourly energy modeling of the building connected to a GHX is critical to the development of a ground source heat pump (GSHP) system that is cost-effective to build, performs efficiently and predictably for the life of the building and is easily maintained. Energy loads are determined by simulating building performance based on the architectural and mechanical system design, anticipated occupancy, and the historical hourly weather data for the location.

Working with the architectural and mechanical design team during the early stages provides the opportunity to adjust energy loads to and from the GHX. Using the energy model to determine the impact of changes to glass specifications, lighting and lighting controls, ventilation air strategy, domestic hot water production, distribution system design, etc. can result in both a reduction in the size and cost of the GHX and an improvement in the efficiency and performance of the system. GEOptimize will work with BR+A to develop several iterations of the hourly energy model and the impact of the changes on the size, cost, and performance of the GHX. GEOptimize will work with BR+A on an hourly basis during the conceptual design

to ensure the client receives a cost effective and sustainable GSHP system. GEOptimize employees will track their hours and update BR+A weekly.

NY state has a thorough water well database that provides details of the stratigraphy of boreholes from water wells drilled in previous years. This provides the information needed to accurately estimate the thermal conductivity and diffusivity of a vertical borehole during the conceptual design of the GSHP system. This stratigraphy plus the land area available for construction of the GHX provides the information needed to calculate the size and performance of the GHX with reasonable accuracy. Conducting a sensitivity analysis based on the anticipated maximum and minimum thermal properties allows us to estimate the expected range of cost if a GSHP system is installed.

After the decision is made by the client to proceed with the installation of a GSHP system and the type of GHX to install, the appropriate conductivity test will be used to confirm the assumptions made during the feasibility study. This approach helps reduce the cost of conducting a conductivity test unnecessarily.

A conductivity test for a vertical GHX is conducted by drilling a test borehole on the site. The depth and location of the test borehole is determined after:

- A review of geological and geotechnical information available for the site and area to estimate the thermal conductivity that can be expected at various depths
- Interviews with drilling contractors experienced in the construction of a vertical GHX in the area to determine the most cost-effective borehole depth based on the anticipated geological conditions
- A review of the site to determine the land area available for construction of the required GHX field
- The size of U-tube piping specified for the test borehole is based on the final energy model iteration and flow rate requirements through the GHX field, the number and depth of boreholes anticipated, and the heat transfer fluid specified for the system. This is required to optimize the capacity of the GHX circulation pump.
- The location of the test borehole will be located for integration into the final GHX field design.

The cost of conducting the thermal conductivity test for a vertical borehole will be based on the depth of borehole specified plus the cost of performing the thermal conductivity test. GEOptimize will coordinate the thermal conductivity test, analyze the data, and generate a report based the test.

Phase 2: Detailed GHX Design

If the feasibility assessment for construction of a GSHP system is acceptable to the client, GEOptimize will:

- Prepare design drawings and specifications for authority approval and tender with BR+A up to and including the GHX manifold.
- Review bid documents and provide construction administration and assist in testing and commissioning of the system. GEOptimize will develop a quality assurance / quality control program with the design team to ensure the system is constructed as designed.
- Coordinate with BR+A on the detailed design of the heat pump system in the facility.

Scope of work

Phase 1: Feasibility of GSHP System

GEOptimize scope of work includes:

- Iterative ground heat exchanger (GHX) model working in collaboration with the engineering design team to optimize the cost and performance of the GHX.

- Preliminary GHX design indicating approximate land area required for the building.
- Work with BR+A to develop budget construction cost estimates for installation of GHX.
- Recommendations for mechanical system optimization with GHX.

Phase 2: Detailed GHX Design

Details of the GHX provided will include:

- Layout of GHX circuit piping for the recommended GHX configuration.
- Design of fusion welded headers to connect GHX circuits to supply / return runout piping.
- Supply / return runout piping from GHX headers to manifold in the building (or vault, if required)
- Written specifications for GHX piping for circuits, headers, and supply / return runout piping.
- Specifications for heat transfer fluid and estimated heat transfer fluid volume.
- GHX manifold design up to isolation valves between interior heat pump system piping and GHX.
- Pressure-drop calculations for the GHX and recommended circulation pump sizing and control.
- Detailed stamped construction drawings of proposed GHX, including layout on building site plan, header design, borehole design, supply / return runout pairs and manifold design.
- Coordination with BR+A in development of mechanical system schematic for interior mechanical system.
- Interview contractors submitting tenders for the project and providing recommendations for contractor.
- Development of QA / QC program for installation of GHX.
- Pre-construction meeting with selected contractor, project site supervisor, mechanical contractor, and other stakeholders in the project to avoid duplication and ensure coordination of installation.
- Site visit during construction of GHX.
- Site visit during commissioning of GHX.
- Develop detailed operating and maintenance manual for GHX.

Pricing

Description	Rate	Qty	Line Total
Phase 1: Feasibility	(Hourly) See rate structure below	1	---
Thermal Conductivity Test specifications and test results analysis. (NOTE: Test borehole and conduction of the thermal conductivity test by others)	\$4,500.00	1	\$4,500.00
Phase 2: Detailed GHX Design as outlined in scope of work	\$174,500.00	1	\$174,500.00
Subtotal (USD)			\$179,000.00

Hourly Rate Structure

Employee	Rate (USD/hr)
Ed Lohrenz	250.00
Ian Lohrenz	225.00
Stephen Hak	200.00
Connor Dacquay	225.00

Terms

- Additional work out of scope will be charged at hourly rates.
- Invoices will be submitted monthly for work completed to date.
- Payment is expected on receipt of invoice.

Authorized signature